

A data-first environmental screening framework identifies urinary MCOP as a robust biomarker signal associated with prevalent colorectal cancer

Abstract

Background: Environmental epidemiology often begins with a preselected exposure, making it difficult to separate biological plausibility from candidate-selection effects. We developed an outcome-blinded, data-first framework that moved from broad toxicogenomic nomination to human biomarker testing and independent colorectal cancer (CRC) disease-state transcriptomics.

Methods: We evaluated 267 environmental chemicals using human chemical-gene associations and CRC-related genes. Before candidate-specific CRC results were inspected, chemicals were filtered using prespecified rules for chemical identity, exposure interpretability, human biomarker availability, analytical detectability, NHANES coverage, and survey-model testability. Eligible chemical-biomarker mappings were reduced to unique NHANES biomarker tests before multiplicity-controlled CRC screening. FDR-supported tests underwent the same robustness audit. Independent CRC transcriptomic datasets were then used to assess whether the nominated biological axis was also altered in the CRC epithelial disease state.

Results: Prespecified actionability filtering reduced 267 candidates to 87 human-testable chemical-biomarker mappings, representing 15 unique NHANES biomarker tests. Two tests passed BH-FDR <0.05: urinary MCOP and serum PFHS. MCOP was the only biomarker meeting the prespecified highest robustness tier. In 9,936 NHANES participants, including 70 prevalent CRC cases, each doubling of urinary MCOP was associated with higher odds of prevalent CRC (OR 1.25, 95% CI 1.08-1.44; P=0.003). Leave-one-cycle-out estimates remained positive with confidence intervals excluding 1, although cycle-specific effects were heterogeneous. Independent CRC epithelial analyses showed reproducible PPAR/nuclear-receptor remodeling across multiple definitions, with lower activity in enterocyte-like tumor epithelium but modestly higher activity in secretory-like tumor epithelium.

Conclusions: An outcome-blinded data-first screen identified urinary MCOP as the only biomarker reaching the highest prespecified robustness tier for association with prevalent CRC. The association is cross-sectional and does not establish a parent-compound-specific or causal relationship. CRC transcriptomics independently implicate a state-specific PPAR/nuclear-receptor axis, providing biological convergence but not evidence of mediation.

Keywords: *colorectal cancer; MCOP; DINP; NHANES; environmental exposure; toxicogenomics; PPAR; biomarker screening*

Introduction

Colorectal cancer (CRC) arises through interactions among inherited susceptibility, aging, lifestyle, metabolic factors, and environmental exposures [1,2]. Environmental chemicals are plausible contributors because many are ubiquitous, measurable in human biomonitoring programs, and capable of altering receptor, transcriptional, or metabolic biology. However, human evidence for individual chemicals remains uneven, and the literature is often organized around exposures selected in advance rather than around a transparent comparison with alternative candidates [3-7].

This creates a candidate-selection problem. In a conventional exposure-first study, the disease association is evaluated only after one chemical or chemical class has already been chosen from a much larger universe. A data-first strategy can separate these steps by requiring candidate nomination and human testability to be defined before candidate-specific disease results are inspected [29,30]. Such separation is especially important when multiple chemicals map to the same measured biomarker, because biomarker availability does not imply chemical equivalence or parent-specific attribution.

A second gap lies between toxicogenomic relevance and human measurability. A chemical can appear biologically relevant yet be poorly detectable, sparsely measured, or represented only through a non-specific biomarker. Conversely, a measurable biomarker may not itself be the molecular entity implicated by

toxicogenomic databases. Translating from molecular nomination to a population-level test therefore requires an explicit actionability layer that evaluates identity, exposure interpretation, measurement performance, and survey feasibility independently of the disease outcome.

We developed an outcome-blinded multistage framework to address these problems. The framework first screened a broad environmental chemical universe for CRC-related toxicogenomic relevance, then applied human actionability rules without using candidate-specific CRC statistics, and only afterward tested the resulting unique NHANES biomarker models. FDR-supported signals were evaluated under a common robustness audit. Finally, independent CRC transcriptomic data were used to ask whether the biological axis implicated by the nominated exposure signal was also remodeled in the human CRC disease state. In this study, "convergence" denotes independent implication of the same broad biological pathway or regulatory axis across toxicologic nomination and CRC disease-state analyses; it does not require concordant direction and does not imply mediation or causal linkage.

Methods

Study design

The analysis comprised five stages: toxicogenomic discovery, outcome-blinded actionability prioritization, systematic NHANES biomarker screening, robustness assessment of supported signals, and independent CRC transcriptomic disease-state analysis. The analytical unit changed across stages from chemical entity, to chemical-biomarker mapping, to unique human biomarker test. These objects were not treated as interchangeable.

Outcome firewall and analysis freeze

The principal design safeguard was an outcome firewall. Before candidate-specific CRC results were inspected, candidate eligibility and biomarker selection were based only on chemical identity, interpretable human exposure mapping, biomarker availability, biomarker detectability, NHANES cycle coverage, and predefined survey-model testability. No CRC-derived statistic was permitted to alter candidate eligibility, biomarker selection, or the number of biomarker tests entering multiplicity correction.

After the outcome firewall was opened, odds ratios, confidence intervals, P values, BH-FDR values, cycle-specific effects, and robustness results were evaluated. The analysis set and manuscript-facing results were subsequently frozen before manuscript construction. We use "outcome-blinded" for the pre-outcome actionability stage and "frozen before manuscript construction" for later analyses; we do not describe the study as prospectively preregistered.

Toxicogenomic discovery

Human chemical-gene associations were obtained from the Comparative Toxicogenomics Database (CTD) [8], and CRC-related genes were obtained from GeneCards [9]. CTD records were interpreted as database associations rather than validated causal targets. Chemical-level enrichment was evaluated using Fisher/hypergeometric testing with multiple-testing correction, with degree-matched permutation used as a sensitivity analysis to reduce the influence of highly connected genes. This stage nominated molecular candidates but did not determine human testability.

Outcome-blinded human actionability

Each of the 267 starting chemicals was evaluated for valid chemical identity, interpretable exposure mapping, availability of a human biomarker, analytical detectability, NHANES cycle coverage, and feasibility of a survey-weighted CRC model. Detectability, coverage, and model-testability thresholds were defined before candidate-specific CRC effects were used. Full gate definitions and candidate-level audit fields are provided in the Supplementary Methods and Supplementary Table S8.

When several chemicals mapped to the same measured biomarker, the chemical-biomarker mappings were retained for audit but collapsed to one biomarker test for human screening. Thus, 87 eligible chemical-biomarker mappings represented 15 unique NHANES biomarker tests. Shared biomarker mapping was not

interpreted as biological equivalence among parent chemicals.

Systematic NHANES biomarker screen

All 15 unique biomarker tests entered the same systematic CRC screen. Survey-weighted logistic regression modeled prevalent CRC as a function of log2-transformed exposure, age, sex, race/ethnicity, body mass index, smoking, and poverty-income ratio; urinary biomarkers additionally included log2 urinary creatinine [11,12,31,32]. BH-FDR was controlled across all 15 tests [10]. FDR-supported tests, rather than a single "leading" result, proceeded to the same robustness audit.

Primary MCOP analysis

The frozen MCOP analysis used NHANES 2005-2006 through 2017-2018. Adults aged ≥ 20 years were included, with prevalent CRC cases compared with cancer-free controls. The complete-case sample contained 9,936 participants and 70 CRC cases. The exposure was log2-transformed urinary MCOP (URXCOP), interpreted as the odds ratio per doubling. Complex-survey inference used cycle-specific phthalate subsample weights divided by seven and cycle-unique strata and primary sampling unit identifiers. R `survey::svyglm` was the manuscript estimator; an independent Taylor-sandwich implementation was used as numerical quality control.

Robustness audit

FDR-supported signals were evaluated under a common robustness framework covering statistical support, leave-one-cycle-out stability, cycle-level behavior, measurement adequacy, exposure robustness, sensitivity to alternative specifications, and analytic reproducibility. The full scoring rubric and fingerprints are reported only in the Supplementary Materials.

For MCOP, frozen sensitivity analyses included restriction to age ≥ 40 years, exclusion of recent CRC diagnoses, exclusion of the upper exposure tail, alternative creatinine handling, pairwise co-exposure adjustment, sex-specific analyses with formal interaction testing, and exposure-shape analyses. Leave-one-cycle-out analysis assessed dependence on any single NHANES cycle, whereas the exposure-by-cycle interaction assessed equality of cycle-specific effects; these were treated as distinct statistical questions.

CRC transcriptomic disease-state analysis

Independent CRC transcriptomic analyses were used to evaluate disease-state remodeling of the PPAR/nuclear-receptor axis. These datasets did not contain DINP or MCOP perturbation and therefore could not test exposure-to-state causality. Main inference used paired patient or donor comparisons rather than treating individual cells as independent replicates. The frozen seven-gene PPAR/nuclear-receptor core was tested alongside independent receptor, pathway, metabolic, and regulon definitions.

Within-state analyses were restricted to source-supported epithelial annotations with adequate paired cell counts. Donor-level delta correlations were used to assess relationships between PPAR/nuclear-receptor remodeling and differentiation, inflammatory, stress, and regulatory programs. These analyses were interpreted as disease-state correspondence rather than mediation.

Results

Outcome-blinded prioritization yielded 15 unique human biomarker tests

Prespecified actionability filtering reduced the 267 candidate chemicals to 87 eligible chemical-biomarker mappings, representing 15 unique NHANES biomarker tests (Fig. 2). The largest attrition occurred during exposure interpretation and human-testability assessment; full stage-specific counts are shown in Fig. 2 and Supplementary Tables S1 and S8.

Within the DINP-related set, MiNP remained a molecular nominee but had insufficient direct urinary detectability for primary human testing in the updated audit. MCOP (URXCOP), a distinct urinary biomarker commonly used to index DINP exposure [13-16], met the measurement and coverage requirements and

therefore entered the systematic human screen. This transition reflected biomarker actionability, not chemical equivalence among MiNP, MCOP, and the DINP parent.

Two biomarker tests passed FDR control, but only MCOP reached the highest robustness tier

Two of the 15 biomarker tests passed BH-FDR <0.05: urinary MCOP showed a positive association with prevalent CRC, whereas serum PFHS showed an inverse association (Fig. 3). The remaining 13 tests did not pass the 15-test FDR threshold.

Both FDR-supported signals underwent the same prespecified robustness audit. MCOP was the only biomarker meeting the highest robustness tier. PFHS remained a supported Tier B signal because its analyzable CRC event count was substantially smaller (approximately 32 cases) and its audit retained a persistent technical/fit warning, preventing it from satisfying the highest-tier analytic criterion [18-21]. The complete rubric and signal-level audit are provided in Supplementary Table S4.

Urinary MCOP was positively associated with prevalent CRC

In the frozen primary survey-weighted model, each doubling of urinary MCOP was associated with higher odds of prevalent CRC (OR 1.25, 95% CI 1.08-1.44; P=0.003) among 9,936 participants, including 70 CRC cases (Fig. 3). An independent Taylor-sandwich implementation reproduced the estimate to numerical precision. Because CRC was prevalent and MCOP was measured at examination, the analysis estimates a cross-sectional biomarker-disease association rather than prospective risk.

MCOP was stable to cycle omission despite significant temporal heterogeneity

All seven leave-one-cycle-out pooled estimates remained above 1 and their 95% confidence intervals excluded 1, indicating that the pooled association was not dependent on a single NHANES cycle (Fig. 4). Prespecified age, diagnosis-timing, upper-tail, creatinine-handling, and co-exposure analyses also retained the positive direction.

Cycle-specific effects were nevertheless heterogeneous: the MCOP-by-cycle interaction was significant (P approximately 0.006), with 2011-2012 the major discordant cycle. Leave-one-cycle-out stability and effect homogeneity address different questions; the former supports robustness of the pooled estimate to cycle omission, whereas the latter shows that effect magnitude was not constant across calendar periods. The source of the heterogeneity cannot be resolved with the present data and may reflect temporal exposure differences, residual population differences, measurement variation, or chance.

Secondary categorical and spline analyses did not provide evidence sufficient to replace the prespecified continuous log2 model and are therefore presented as secondary characterization in the Supplementary Materials.

CRC showed epithelial-state-specific PPAR/nuclear-receptor remodeling

Independent CRC epithelial analyses demonstrated robust remodeling of the PPAR/nuclear-receptor axis (Fig. 5). The frozen seven-gene core was lower in tumor-derived epithelium (median tumor-normal delta -0.419; FDR approximately 9.3×10^{-7}), and most independent receptor, KEGG, Reactome, metabolic, and regulon definitions showed concordant down-remodeling at the aggregate epithelial level.

The effect was not adequately described as global PPAR suppression. Within enterocyte-like epithelium, the PPAR/nuclear-receptor score was lower in tumor samples (n=24 paired donors; median delta -0.191; FDR 5.3×10^{-4}), whereas secretory-like epithelium showed a modest increase (n=27; median delta +0.068; FDR=0.018). Together with epithelial compositional redistribution, these findings support state-specific regulatory remodeling rather than a uniform pan-epithelial effect [27,28,36-38].

Donor-level analyses did not support a simple PPAR-low to RELA/STAT3-high mediation model. Toxicologic evidence and CRC transcriptomics therefore intersect at a broad nuclear-receptor/PPAR biological axis, but the exposure-to-epithelial-state link remains untested.

Discussion

This study asked whether an outcome-blinded, data-first framework could move from a broad environmental chemical universe to a human biomarker signal associated with CRC while keeping candidate selection auditable. The framework reduced 267 candidates to 15 unique NHANES biomarker tests, identified two FDR-supported signals, and classified urinary MCOP as the only biomarker meeting the highest prespecified robustness tier. The principal contribution is therefore not a claim that DINP causes CRC, but a transparent discovery architecture that separates chemical nomination, human measurability, population association, and downstream biological interpretation.

Within NHANES, the MCOP association was credible on several internal dimensions. It survived multiplicity correction in the 15-test screen, remained positive under a common robustness audit, was reproduced numerically across independent survey implementations, and was not eliminated by omission of any single cycle. At the same time, the significant exposure-by-cycle interaction shows that robustness to cycle omission is not equivalent to effect homogeneity. The pooled association is therefore best viewed as internally robust but temporally heterogeneous, rather than as seven homogeneous replications.

The epidemiologic result does not establish temporal or causal risk. CRC was prevalent, urinary MCOP was measured at examination, and only 70 complete-case CRC cases contributed to the primary model. Reverse causation, survivor effects, short-term variability in a spot urinary biomarker [33-35], residual confounding, and calendar-period heterogeneity remain plausible limitations. These constraints are central to interpretation but do not invalidate the narrower conclusion that a systematic screen identified a reproducible cross-sectional MCOP-CRC signal within the available NHANES data.

The transcriptomic analyses add an independent disease-state layer. PPAR/nuclear-receptor remodeling was reproducible across multiple pathway and regulon definitions, but the within-state audit changed the biological interpretation: enterocyte-like epithelium showed suppression, whereas secretory-like epithelium showed modest elevation. Thus, the CRC phenotype is more accurately described as state-specific PPAR/nuclear-receptor remodeling than as global pathway suppression. This distinction is important because it prevents a complex epithelial state transition from being reduced to a single pathway-direction claim.

Previous studies have reported conflicting roles and expression patterns for PPARs in colorectal cancer, with both the direction and functional interpretation varying across PPAR isoforms, experimental models, and disease contexts [22-26]. Our epithelial-state-resolved analysis offers a potential explanation for part of this apparent inconsistency: the PPAR/nuclear-receptor axis was not uniformly suppressed across CRC epithelium, but instead differed between enterocyte-like and secretory-like states. Accordingly, bulk or aggregate measurements may obscure opposing state-specific patterns. These findings may help reconcile part of the previously discordant PPAR literature in CRC, while not implying that epithelial state composition explains all historical discrepancies.

The toxicologic and transcriptomic layers show a biologically plausible but untested intersection. MiNP contributed to nomination of a DINP-related molecular axis, while MCOP provided the viable human biomarker test; these entities remain chemically distinct. Independent CRC transcriptomics implicate the same broad nuclear-receptor/PPAR biology, but the present data do not demonstrate that DINP or MCOP induces the observed epithelial state or that PPAR remodeling mediates the epidemiologic association. Direct intestinal epithelial or organoid perturbation experiments are required to test that bridge.

The study's principal strength is design transparency rather than causal depth. Candidate-selection rules were separated from CRC outcome statistics, the multiplicity denominator was defined at the unique biomarker-test level, competing FDR-supported signals were subjected to the same robustness audit, and disease-state transcriptomics were interpreted independently of the exposure association. The next decisive steps are prospective or prediagnostic replication of the MCOP-CRC association and controlled intestinal epithelial experiments that test whether DINP-related exposure can reproduce the state-specific nuclear-receptor remodeling observed in CRC.

Conclusion

The data-first screen identified urinary MCOP as the only biomarker meeting the prespecified highest robustness tier for association with prevalent CRC. MCOP is commonly used to index DINP exposure, but the present study does not establish a parent-compound-specific or causal relationship. Independent CRC transcriptomics demonstrate epithelial-state-specific PPAR/nuclear-receptor remodeling, providing biological convergence while leaving the environmental-to-epithelial causal bridge open for direct testing.

Figure Legends

Figure 1. Study framework: data-first discovery, outcome firewall, human screening, robustness audit, and independent disease-state analysis.

Figure 2. Outcome-blinded actionability defines the human screening universe. (A) Prespecified gates reduced 267 core environmental chemicals to 87 human-testable chemical–biomarker mappings. CRC outcome statistics remained behind the outcome firewall until the biomarker-test universe was frozen; 27 mappings met the strict D2/C2/T2 rule. (B) The 87 eligible mappings represented 15 unique NHANES biomarker tests, which formed the denominator for BH-FDR correction. Point area reflects the number of eligible chemical mappings and color denotes biological matrix. (C) MiNP, parent DINP and urinary MCOP retained distinct roles. MiNP was a molecular nominee but failed the direct-detectability gate, parent DINP was not a significant Phase 1 hit, and MCOP entered the human screen as a measurable biomarker for a DINP-related exposure axis. Biomarker translation does not imply chemical equivalence or a direct MCOP molecular hit.

Figure 3. Complete 15-test systematic human screen, FDR-supported signals (MCOP and PFHS), and the primary MCOP estimate.

Figure 4. MCOP robustness evaluation: leave-one-cycle-out stability, sensitivity analyses, cycle heterogeneity, and secondary exposure-shape characterization.

Figure 5. CRC disease-state biology: state-specific PPAR/nuclear-receptor remodeling, highlighting opposing epithelial-state patterns that may help explain prior inconsistent PPAR findings. The environmental bridge is dashed and explicitly untested.

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